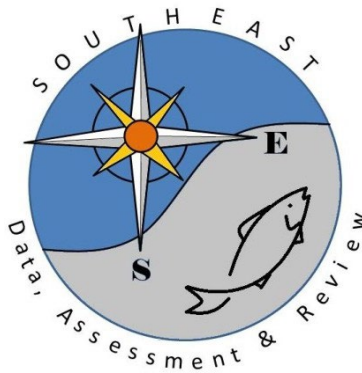




SEDAR

Southeast Data, Assessment, and Review

SEDAR 87

Gulf White, Pink, and Brown Shrimp

SECTION IV: Research Recommendations

SEDAR
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1. DATA WORKSHOP RESEARCH RECOMMENDATIONS

1.1 COMMERCIAL FISHERY STATISTICS RESEARCH RECOMMENDATIONS

- Continue investigations into the methodology of the GSS to better understand differences between GSS and STT programs where they overlap.
- Improve effort data collection for inshore shrimping trips.
- Develop a method to quantify under-reporting of landings in the shrimp fishery, perhaps through the use of separate socio-economic surveys.
- Quantify the prevalence of misidentification of “hopper” brown shrimp within each Gulf state.
- Continue investigations into estimation of species-specific effort.

1.2 ENVIRONMENTAL DRIVERS/INDUSTRY REPORT RESEARCH RECOMMENDATIONS

We highlight data gaps and research recommendations that will improve understanding of environmental impacts on shrimp.

Updated life history information – Much of the life history obtained for shrimp was conducted in the 1960-1970s. Given the short generation time of shrimp, high environmental influence on productivity, and nonstationary environmental dynamics, life history dynamics have likely changed, and these parameters should be updated (e.g., SEDAR87-DW-05).

Density dependence - The suggestion to prioritize environmental impacts on the estuarine stages of shrimp relies on the assumption that this is the region in which density dependence occurs. Expert guidance has suggested that there is no relationship between post larvae entering the estuary and the number of spawning adults in the population (J Nance). While previous efforts suggest that density dependence occurs during estuary residence (e.g., Galveston Bay sub-adult survey to predict brown fishery dynamics for the upcoming year), further research to support this assumption would further validate the assumptions made herein.

Population connectivity models for brown and white shrimp – Research linking spawning adults to their nursery estuary may provide guidance on which estuarine habitats are most productive. Priority could then be given to these most influential nursery habitats for research and conservation.

Mechanistic environmental relationships – Hypothesized environmental drivers presented in this report are correlative only and do not attempt to identify the mechanistic relationship underlying these correlations. Further research identifying the exact driver and the organismal and stock-wide response to these drivers would improve this effort. Updated environmental relationships should be generated or explored.

State FIM data standardization - Additional research should be conducted to standardize and calibrate state-by-state surveys.

Role of shrimp as key forage species - As forage species, shrimp play a key role in the ecosystem. Additional research may clarify linkages between shrimp and predator species and better clarify the extent of predator-induced mortality (e.g., Fujiwara et al. 2016). These linkages may inform how these interrelated species should be managed from an ecosystem-based perspective.

1.3 MEASURES OF POPULATION ABUNDANCE RESEARCH RECOMMENDATIONS

- Explore survey / gear calibration studies among state and federal sampling programs
- Perform post hoc analysis to potentially account for habitat classification variables and on indices of abundance
- Examination of whether due to zeros, indices based on monthly data may best be structured to focus on core recruitment months or accommodate in model
- Exploration of indices of abundance utilizing combined data from AL, MS, and LA 16 ft state sampling programs, including potentially including a weighting factor to account for differences in area sampled (surface area, habitat area, etc.)

1.4 ECONOMICS AND SOCIAL SCIENCES

No research recommendations were provided.

2. ASSESSMENT PROCESS RESEARCH RECOMMENDATIONS

2.1 White Shrimp

The models provided in this report are sufficient to provide management advice for the stock. However, should future research funding become available, we have provided suggestions below.

Potential improvements to the modeling framework include accounting for removal of shrimp as it pertains to harvest rates that are optimized at varying size classes. Creating a feedback loop that appropriately represents the removal of larger shrimp that may not contribute to future generations as well as the removal of smaller shrimp that may not grow into large shrimp should be accounted for. Sensitivities of these potential feedback loops and their impact on estimating optimal harvest rates should be investigated in both directions (i.e. Large to Small and Small to Large impacts).

Additional research into covariates may also be investigated. Direct inclusion of covariates generally resulted in improved model fits and could likely improve forecasting efficiency for trends of abundance. To forecast MSY, covariates would need to be projected into the future. For environmental covariates, the cyclical nature of these trends would need to be captured. For economic covariates, the relationship with projected harvest rates would need to be explicitly defined.

Implications of the LDWF survey capturing mostly small shrimp in conjunction with the fishery capturing mostly large shrimp should be investigated as it pertains to catchability estimation and resulting estimates of escapement. It is suspected that this was the primary driver for a large process variance in the model.

2.2 *Pink Shrimp*

Additional research into covariates should be investigated. Direct inclusion of covariates resulted in improved model fits and could likely improve forecasting efficiency for trends of abundance. To forecast MSY, covariates would need to be projected into the future. For environmental covariates, the cyclical nature of these trends would need to be captured. For economic covariates, the relationship with projected harvest rates would need to be explicitly defined.

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As funding for scientific surveys is becoming increasingly sparse, implications of using an average of 2019/21 for missing summer 2020 SEAMAP data and resulting effects on model diagnostics should be investigated. EDM performs best on continuous, long time series of data, and quantifying implications of future gaps in survey data would be valuable.

2.3 *Brown Shrimp*

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Potential improvements to the modeling framework include accounting for removal of shrimp as it pertains to harvest rates that are optimized at varying size classes. Creating a feedback loop that appropriately represents the removal of larger shrimp that may not contribute to future generations as well as the removal of smaller shrimp that may not grow into large shrimp should be accounted for. Sensitivities of these potential feedback loops and their impact on estimating optimal harvest rates should be investigated in both directions (i.e. Large to Small and Small to Large impacts).

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categories. Instead of reporting raw R^2 values, using an adjusted R^2 could improve model comparison by penalizing model complexity (e.g., number of embedding dimensions $\times$ size bins $\times$ seasons):

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1-R^2) \cdot (n-1)}{n-k-1} \right)$$

where k is the number of lags combined with any additional parameters used.

Track q and other metrics through the addition of each data point of escapement within EDM. It would be useful to visualize the sample size of escapement necessary for q (or other metrics) to stabilize.

It was difficult to know if data series were long enough to reliably fit EDM models despite some recommendations in the literature. It would be useful to fit EDM models to long simulated or real data sets sequentially eliminating the first point in the data set so that sequential model runs are based on shorter and shorter time series. The performance or stability of the model will probably degrade as the time series becomes too short. Such tests could be done routinely in real stock assessments or to develop general rules.

Can scale be estimated in EDM when survey data are flat or constitute a one-way trip? Such survey data patterns are problematic and degrade accuracy using traditional assessment models.

JABBA does not directly utilize environmental, economic or social data. However, a similar Bayesian surplus production model could be programmed locally to use such information.

Consider incorporating bay trawls from both Texas and Louisiana (with their accompanying environmental collections) to create a recruitment and adult index into the EDM model for Brown Shrimp.

If EDM can handle environmental data that has an unspecified probably nonlinear relationship with stock size, it seems possible that it might handle fishery CPUE as well. Fishery-dependent CPUE is meaningful to constituents but hard to handle in assessment models because it has a nonlinear relationship with biomass. It usually changes more slowly than biomass when stock size is high and more slowly than stock size when biomass is low. Moreover, CPUE is affected by changes in fishing technology, and vessel operations. One advantage of CPUE is that it can be standardized in models like VAST to account for differences among vessels, locations, etc. in the same way as survey data (although nonlinearity is not removed). CPUE data are usually less

variable than survey data and may therefore show the direction, if not the magnitude, of changes in stock size clearly. The possibility of using CPUE in EDM should be investigated.

It might be possible to use fishery dependent data to a greater extent to improve short-term projections with EDM. EDM makes predictions based on lagged data which are “nearby” in time (e.g. 3-4 year lags). The emphasis on recent data may alleviate problems using fishery dependent data with properties that change slowly over time. One year of missing survey or other predictor data have a disproportionate effect on the length of time series in EDM models because each datum is used repeatedly in lags. One year of missing survey data for brown shrimp was imputed by averaging the previous and subsequent observation. Another imputation approach may be better, e.g., based on a nonlinear fit to data for the same spatial cell and population in years $t-3$... $t-1$ and $t+1$... $t+3$.

Use simulation, bootstrapping or some other approach to determine if the variances for EDM model results are realistic.

Determine if omitted predictor variables in EDM bias the scale of stock size estimates. According to Takens' theorem, the trend in EDM model estimates may be insensitive to missing predictor variables because the lagged survey data used in the model were influenced by and carry information about effects of missing predictors, particularly if predictor variables are correlated. However, it is not clear how or if Takens' theorem applies to scale.

Correlated predictor variables are a significant problem in traditional modeling (e.g., with linear regression). Evaluate effects of correlated predictors on EDM models. Would it be better to avoid correlated variables or use, for example, uncorrelated principal component scores that carry the same information as the original data? If correlated predictors cause problems and environmental variables are correlated with the survey data in the model, is it better to avoid using the environmental variables?

Determine how well EDM models can be used to estimate MSY reference points if data were collected while the stock was not near B_{msy} or K . Is it possible to estimate these parameters in EDM for a lightly or chronically overfished stock? That is, can models be extrapolated into unsampled areas of the data space?

Clarify the interpretation of the q parameter in EDM. The parameter is estimated in the EDM model where it is used to put catch and survey data on the same scale, then to scale

“escapement” into biomass units (see Fig. 1). The current EDM formulation is an algebraic reordering of terms in a crude, approximate surplus model that is most accurate under stable stock conditions when somatic growth (G) and mortality rates (Z) are low. However, G and Z are both high for shrimp and the approximation may be suboptimal.

The q parameter is used in EDM to relate biomass during a relatively short survey to biomass at the beginning of the current time step ($I_t = qB_t$, see below). However, when applied to escapement ($I_t - qC_t$), the parameter q converts catch in year t to a hypothetical quantity that adjusts biomass at the beginning of the year as though catch also occurred at the beginning of the year. However, shrimp catch occurs throughout the year while substantial growth and mortality occurs (it is possible that most of the catch biomass is due to growth in the current year). Thus, growth, mortality and the different time scales of the survey and fishery are ignored by the formulation in a manner that might affect model performance for stocks like shrimp with high growth and mortality rates. Other surplus production formulations are available that might serve better for shrimp.

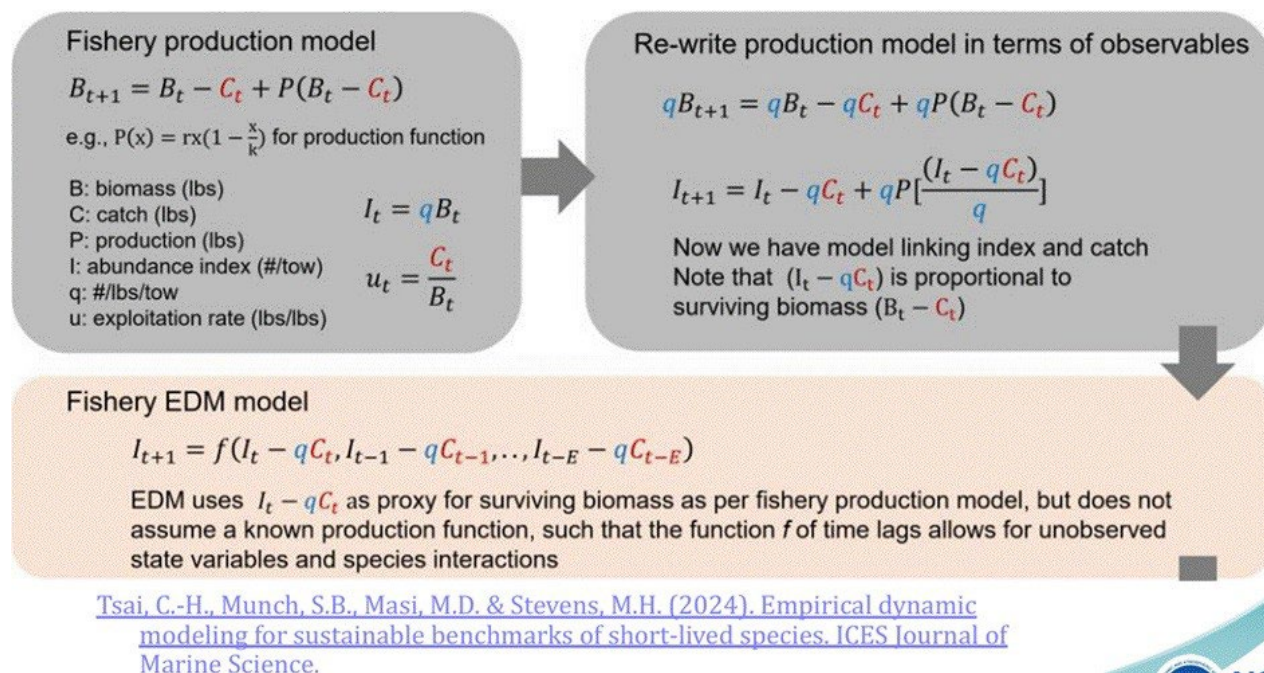


Figure 1. Parameterized Fishery EDM for Gulf shrimps used for this assessment. Taken from Tsai, et al. (2024).

Clarify assumptions in the production model behind EDM. The current formulation (see above) appears to assume that production during year t is a function of biomass at the beginning of the same year t ($B_t - C_t$) with an adjustment for catch occurring later during the same year. The

model subtracts catch in year t from biomass in the same year $B_t - C_t$ again to account for reductions in biomass due to catch. Are we double counting the catch? Would it make sense to write the model as $B_{t+1} = B_t + P(B_t) - C_t$ i.e., assuming production is a function of biomass at the beginning of the year which seems reasonable.

The transformations applied (log, gr1, gr2) could be clarified. If log and gr1 apply only to CPUE or X in the escapement equation, while gr2 transforms the entire escapement term, this distinction should be made explicit. All transformations appear appropriate, q would be estimated to convert catch units to whatever scale is chosen.

This assessment utilized novel approaches for estimating the management quantities of B_{msy} , $B_{current}/B_{msy}$, F_{msy} , $F_{current}/F_{msy}$ for Gulf shrimp, and their trajectories of biomass and fishing rates over the past and projection into the future. While far removed from a “traditional” Research Track Assessment under typical SEDAR operations, these methods proved to be sound as well as robust, at least for brown and white shrimp. The caveat being despite the known relationships between Gulf shrimp species and environmental conditions in the estuaries during their post-larval, juvenile, and sub-adult stages (particularly water temperature in the spring when initial recruitment begins for brown shrimp), none of the environmental data was ultimately included in assessment. The same can be said for the socioeconomic factors that have been shown to drive much of the effort of the fleet (diesel price, consolidation of the fleet, massive recent influx of foreign imports and its effect on dockside prices, etc.). An industry representative stated that “using fishery dependent data in modeling would tend to enhance industry confidence in results, despite difficulties in their use”. The structure of EDM models may provide opportunities in this regard.

Long term predictions based on EDM models with environmental data may not be practical because of uncertainty about future environmental conditions. It may be feasible, however, to carry out short-term predictions. Environmental conditions may be relatively constant or predictable over shorter time periods so that environmental data are easier to predict with some accuracy. Moreover, environmental effects may be implicit in the survey data.